

English
Medium



EDU TERIA

72nd BPSC

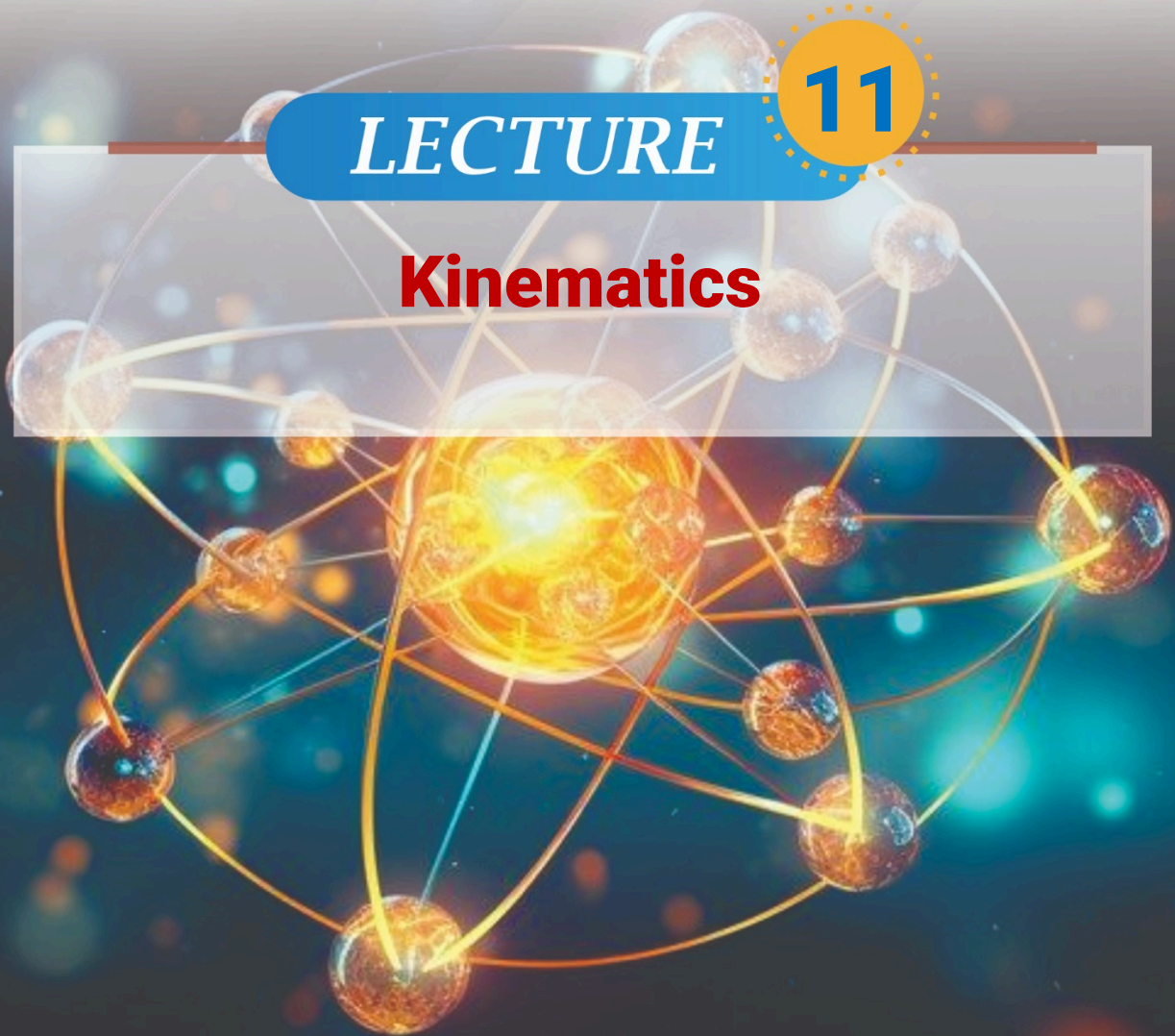
Prelims Online Batch

PHYSICS

LECTURE

11

Kinematics



Kinematics

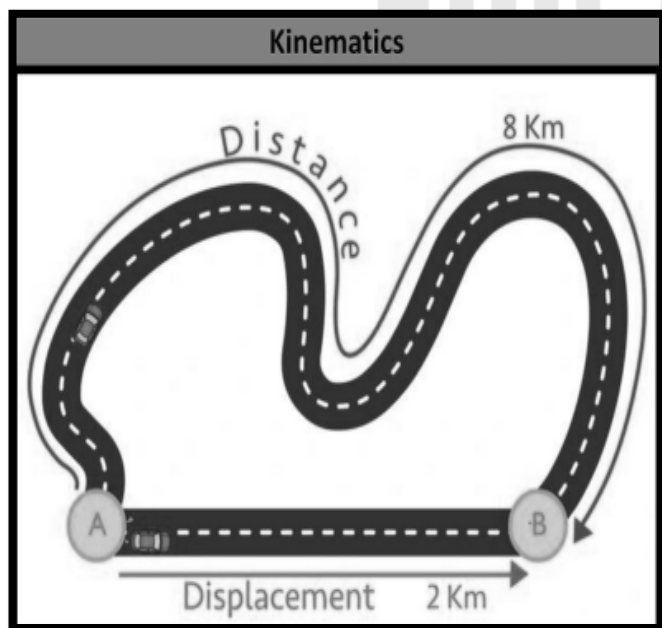
Kinematics

Kinematics is the branch of classical mechanics that describes the motion of points, objects, and systems without considering the forces causing the motion.

- Kinematics is the branch of physics that describes motion using concepts such as displacement, velocity, and acceleration, without considering the forces that cause the motion
- Rest and Motion:** An object is at rest if its position does not change with time relative to its surroundings; otherwise, it is in motion. Rest and motion are relative terms and depend entirely on the chosen frame of reference.

Distance and Displacement:

- Distance is the actual path length covered by an object (a scalar quantity).
- Displacement is the change in position, a vector quantity pointing from the initial to the final position.

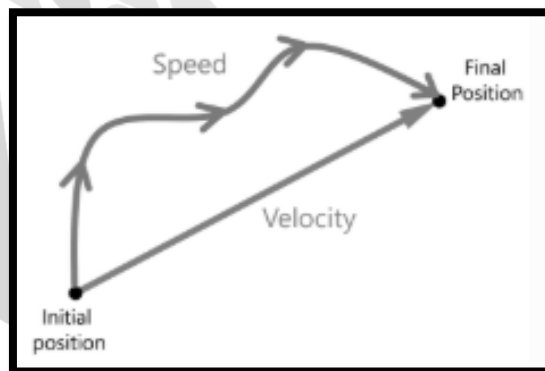


Difference between distance and displacement:

Distance	Displacement
Total length covered between initial and final point known as distance.	Shortest length covered between initial and final point known as displacement.
It has only magnitude. Hence, scalar quantity. E.g.-A person goes 10m	It has magnitude as well as direction also. Hence, known as vector quantity. E.g.- Mohan goes 10m in east direction.
Value of distance will be always positive.	Displacement can be positive, negative and zero.
Distance = speed × time	Displacement = Velocity × time

Speed and Velocity

- Speed is the rate of change of distance (a scalar quantity).

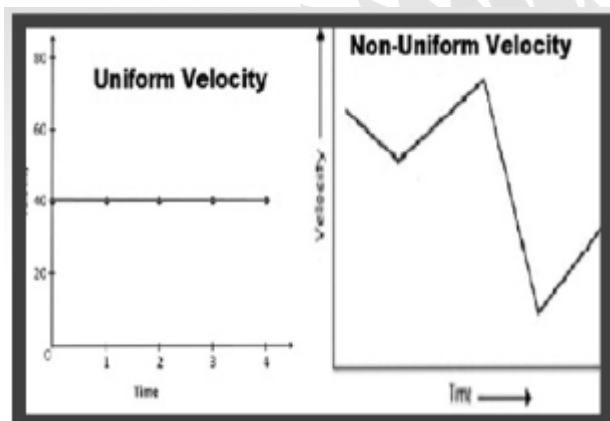


- Velocity is the rate of change of displacement, specifying both magnitude and direction (a vector quantity). A change in either speed or direction results in a change in velocity.
- Rate of change of distance known as speed.
- Rate of change of displacement known as velocity.
- Unequal distance in equal interval of time non-uniform speed.



**Uniform vs. non-Uniform Velocity****Uniform velocity**

- Uniform velocity means that an object moves with constant velocity, i.e., both its speed and direction remain unchanged throughout the motion. In this case, the object covers equal displacements in equal intervals of time in a fixed direction. Since velocity is not changing, the acceleration is zero
- For example, a car moving on a straight road at a constant speed of 60 km/h towards east has uniform velocity. This type of motion is rare in real life because even a slight change in direction or speed makes velocity non-uniform.

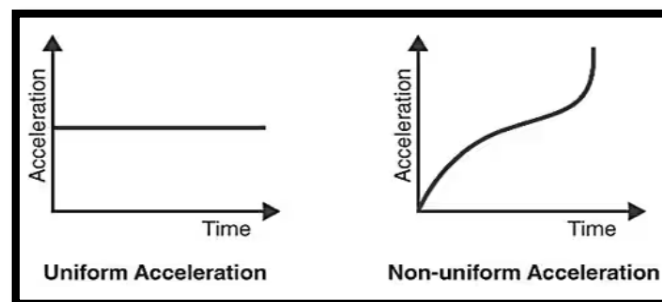


- **Non-uniform velocity**— Non-uniform velocity means that the velocity of an object changes with time. This change can happen due to a variation in speed, direction, or both. In such motion, the object covers unequal displacements in equal intervals of time or changes direction, and therefore, acceleration is present.
- For example, a car moving in traffic (changing speed), or a body moving in a circular path (constant speed but changing direction), both have non-uniform velocity

Uniform acceleration vs. non-uniform acceleration

Uniform acceleration- Uniform acceleration means that the velocity of an object changes by equal amounts in equal intervals of time. In other words, acceleration remains constant throughout the

motion. This implies that the object either increases or decreases its velocity at a steady rate.



For example, a freely falling body under gravity near the Earth's surface experiences uniform acceleration due to Acceleration due to gravity, which is approximately 9.8 m/s^2 (ignoring air resistance). In such motion, the standard equations of motion are applicable.

Non-uniform acceleration — Non-uniform acceleration means that the velocity changes by unequal amounts in equal intervals of time. Here, acceleration is not constant; it varies either in magnitude, direction, or both.

For example, a car moving in city traffic (constantly accelerating and decelerating), or a body moving in a circular path (direction of velocity changes continuously), both exhibit non-uniform acceleration.

Equation of motion

Equations of motion describe the relationship between displacement, velocity, acceleration, and time when an object moves with uniform (constant) acceleration. These equations are fundamental in Kinematics and are widely used in numerical.

Kinematics

- For objects moving with constant acceleration, the following equations relate initial velocity (u), final velocity (v), acceleration (a), displacement (s), and time (t):
- We obtain following equations:

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$\text{Displacement in the } n^{\text{th}} \text{ second : } S_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1)$$

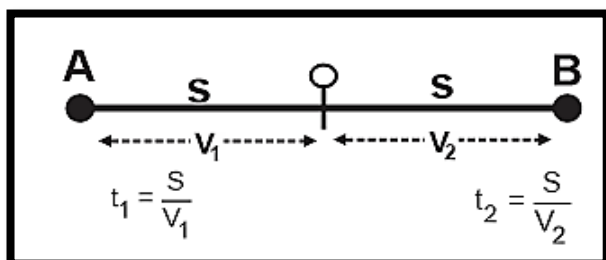


Case- I — If a particle travels the first half distance with speed v_1 and second half distance with speed v_2 . Find its average speed during the journey.

Solution

$$V_{av} = \frac{s + s}{t_1 + t_2} = \frac{2s}{\frac{s}{v_1} + \frac{s}{v_2}} = \frac{2v_1 v_2}{v_1 + v_2}$$

Note :- Here V_{av} is the harmonic mean of two speeds.



Case-II —

If a particle travels with speed v_1 during first half time interval and with speed v_2 during second half time interval. Find its average speed during its journey.

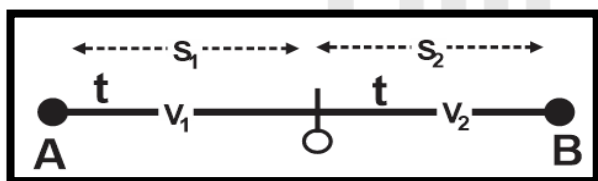
Solution

$$s_1 = v_1 t \text{ and } s_2 = v_2 t$$

$$\text{Total distance} = s_1 + s_2 = (v_1 + v_2)t$$

$$\text{Total time} = t + t = 2t$$

$$\text{then } v_{av} = \frac{s_1 + s_2}{t + t} = \frac{(v_1 + v_2)t}{2t} = \frac{v_1 + v_2}{2}$$



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